PDA03 Module 8.1: Exploring Student Performance Data Analysis Assignment

Task 0: Create the student performance dataset

In [19]:
```
# Not required because it has already been provided

# '''
# # Run the code to generate the file
# np.random.seed(0)
# num_samples = 1000
#
# # Generate random scores between 0 and 100 for each subject
# math_scores = np.random.randint(0, 101, num_samples)
# science_scores = np.random.randint(0, 101, num_samples)
# english_scores = np.random.randint(0, 101, num_samples)
#
# # Create a DataFrame
# df = pd.DataFrame(
#     {
# 'Math': math_scores,
# 'Science': science_scores,
# 'English': english_scores
# })
#
# # Save the DataFrame to a CSV file
# df.to_csv('student_performance.csv', index=False)
# '''
```

Task 1: Import the necessary libraries

In [20]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Task 2: Load the dataset

In [21]:
```
# Use Pandas to read the provided CSV file
df = pd.read_csv('.\\sample_data\\student_scores_2000_rows.csv')
df.head()
```

Out[21]:

	id	first_name	last_name	email	gender	part_time_jo
0	1	Paul	Casey	paul.casey.1@gslingacademy.com	male	Fal
1	2	Danielle	Sandoval	danielle.sandoval.2@gslingacademy.com	female	Fal
2	3	Tina	Andrews	tina.andrews.3@gslingacademy.com	female	Fal
3	4	Tara	Clark	tara.clark.4@gslingacademy.com	female	Fal
4	5	Anthony	Campos	anthony.campos.5@gslingacademy.com	male	Fal

In [22]:

```
# Convert the DataFrame to a NumPy array for numerical analysis
scores_array = df.to_numpy()
scores_array
```

Out[22]:

```
array([[1, 'Paul', 'Casey', ..., 63, 80, 87],
       [2, 'Danielle', 'Sandoval', ..., 90, 88, 90],
       [3, 'Tina', 'Andrews', ..., 65, 77, 94],
       ...,
       [1998, 'Madison', 'Cross', ..., 68, 94, 78],
       [1999, 'Brittany', 'Compton', ..., 95, 88, 75],
       [2000, 'Natalie', 'Smith', ..., 83, 93, 100]],
      shape=(2000, 17), dtype=object)
```

Task 3: Data cleaning

In [23]:

```
# Check for any missing or null values and handle them appropriately
df.isnull().sum()
```

Out[23]:

```
id                           0
first_name                   0
last_name                    0
email                        0
gender                       0
part_time_job                0
absence_days                 0
extracurricular_activities   0
weekly_self_study_hours      0
career_aspiration            0
math_score                   0
history_score                0
physics_score                0
chemistry_score              0
biology_score                0
english_score                0
geography_score              0
dtype: int64
```

In [24]:

```
missing_values = df.isnull().sum().sum()

if missing_values == 0:
    print("*** No missing values found ***")
else:
    print(f"*** {missing_values} missing value(s) found. Please handle appropria
```

```
*** No missing values found ***
```

In [25]:
```
# There are no missing values

# If there were any missing values, we could handle them either by filling them

# We can proceed to ensure all data types are correct
print(df.dtypes)
```

```
id                            int64
first_name                   object
last_name                    object
email                        object
gender                       object
part_time_job                  bool
absence_days                  int64
extracurricular_activities     bool
weekly_self_study_hours       int64
career_aspiration            object
math_score                    int64
history_score                 int64
physics_score                 int64
chemistry_score               int64
biology_score                 int64
english_score                 int64
geography_score               int64
dtype: object
```

In [26]:
```
# Ensure all data types are correct (scores should be numeric)
df_scores = df[[col for col in df.columns if "score" in col]]

if df_scores.dtypes.eq('int64').all():
    print("*** All data types are correct ***")
else:
    print("*** Data type issues found. Please ensure all scores are numeric ***"
```

```
*** All data types are correct ***
```

In [27]:
```
# If there were any non-numeric data types for the scores, we could convert them

# After handling any missing values and non-numeric data types, it would then al
scores_array = df_scores.to_numpy()
scores_array
```

Out[27]:
```
array([[ 73,  81,  93, ...,  63,  80,  87],
       [ 90,  86,  96, ...,  90,  88,  90],
       [ 81,  97,  95, ...,  65,  77,  94],
       ...,
       [ 97,  85,  63, ...,  68,  94,  78],
       [ 51,  96,  72, ...,  95,  88,  75],
       [ 82,  99,  91, ...,  83,  93, 100]], shape=(2000, 7))
```

Task 4: Exploratory data analysis (EDA)

In [28]:
```
# Now the dataset is clean and ready for analysis

# Calculate descriptive statistics for each subject: mean, median, mode, range,

# This can be done using the describe() method of Pandas DataFrame

summary_stats = df_scores.describe()
```

```python
print(summary_stats)

for col in summary_stats.columns:
    mean_val = df_scores[col].mean()
    median_val = df_scores[col].median()
    mode_val = df_scores[col].mode().iloc[0]
    range_val = df_scores[col].max() - df_scores[col].min()
    variance_val = df_scores[col].var()
    std_val = df_scores[col].std()

    print("\n")
    print(f"Descriptive statistics for {col}:")
    print(f"Mean: {mean_val}")
    print(f"Median: {median_val}")
    print(f"Mode: {mode_val}")
    print(f"Range: {range_val}")
    print(f"Variance: {variance_val}")
    print(f"Standard deviation: {std_val}")

# Determine the subject with the highest variance in scores
highest_variance_subject = df_scores.var().idxmax()
print("\n")
print(f"The subject with the highest variance in scores is: {highest_variance_su
      f"\nWith a variance of: {df_scores.var().max()}")
```

```
        math_score  history_score  physics_score  chemistry_score  \
count  2000.000000    2000.000000    2000.000000      2000.000000
mean     83.452000      80.332000      81.336500        79.995000
std      13.224906      12.736046      12.539453        12.777895
min      40.000000      50.000000      50.000000        50.000000
25%      77.000000      69.750000      71.000000        69.000000
50%      87.000000      82.000000      83.000000        81.000000
75%      93.000000      91.000000      92.000000        91.000000
max     100.000000     100.000000     100.000000       100.000000

       biology_score  english_score  geography_score
count     2000.00000    2000.000000      2000.000000
mean        79.58150      81.277500        80.888000
std         13.72219      12.027087        11.637705
min         30.00000      50.000000        60.000000
25%         69.00000      72.000000        71.000000
50%         81.00000      83.000000        81.000000
75%         91.00000      91.000000        91.000000
max        100.00000      99.000000       100.000000


Descriptive statistics for math_score:
Mean: 83.452
Median: 87.0
Mode: 99
Range: 60
Variance: 174.89814507253627
Standard deviation: 13.22490624059529


Descriptive statistics for history_score:
Mean: 80.332
Median: 82.0
Mode: 88
Range: 50
Variance: 162.20687943971987
Standard deviation: 12.736046460331396


Descriptive statistics for physics_score:
Mean: 81.3365
Median: 83.0
Mode: 96
Range: 50
Variance: 157.23788669334667
Standard deviation: 12.539453205516844


Descriptive statistics for chemistry_score:
Mean: 79.995
Median: 81.0
Mode: 94
Range: 50
Variance: 163.27461230615307
Standard deviation: 12.777895456848638


Descriptive statistics for biology_score:
Mean: 79.5815
Median: 81.0
```

```
Mode: 100
Range: 70
Variance: 188.29850700350178
Standard deviation: 13.722190313630758


Descriptive statistics for english_score:
Mean: 81.2775
Median: 83.0
Mode: 90
Range: 49
Variance: 144.65081915957978
Standard deviation: 12.027086894156032


Descriptive statistics for geography_score:
Mean: 80.888
Median: 81.0
Mode: 88
Range: 40
Variance: 135.43617408704353
Standard deviation: 11.6377048461904


The subject with the highest variance in scores is: biology_score
With a variance of: 188.29850700350178
```

In [29]:

```
# Calculate descriptive statistics for each subject: mean, median, mode, range,

# This can also be done using NumPy functions directly on the NumPy array

means = np.mean(scores_array, axis=0)
medians = np.median(scores_array, axis=0)

# NumPy doesn't have a mode function, but SciPy does, and it can operate on a Nu
from scipy import stats
modes = stats.mode(scores_array, axis=0)[0]

ranges = np.ptp(scores_array, axis=0)
variances = np.var(scores_array, axis=0)
stds = np.std(scores_array, axis=0)

desc_stats_dict = {
    'subject': df_scores.columns.to_numpy(),
    'mean': means,
    'median': medians,
    'mode': modes,
    'range':ranges,
    'variance': variances,
    'std': stds}

for i, subject in enumerate(desc_stats_dict['subject']):
    print("\n")
    print(f"Descriptive statistics for {subject}:")
    print(f"Mean: {desc_stats_dict['mean'][i]}")
    print(f"Median: {desc_stats_dict['median'][i]}")
    print(f"Mode: {desc_stats_dict['mode'][i]}")
    print(f"Range: {desc_stats_dict['range'][i]}")
    print(f"Variance: {desc_stats_dict['variance'][i]}")
    print(f"Standard deviation: {desc_stats_dict['std'][i]}")
```

```
# Determine the subject with the highest variance in scores
numpy_highest_variance_subject = desc_stats_dict['subject'][np.argmax(desc_stats
print("\n")
print(f"The subject with the highest variance in scores is: {numpy_highest_varia
      f"\nWith a variance of: {np.max(desc_stats_dict['variance'])}")
```

```
Descriptive statistics for math_score:
Mean: 83.452
Median: 87.0
Mode: 99
Range: 60
Variance: 174.810696
Standard deviation: 13.22159960065347


Descriptive statistics for history_score:
Mean: 80.332
Median: 82.0
Mode: 88
Range: 50
Variance: 162.125776
Standard deviation: 12.73286205061533


Descriptive statistics for physics_score:
Mean: 81.3365
Median: 83.0
Mode: 96
Range: 50
Variance: 157.15926775
Standard deviation: 12.536317950259559


Descriptive statistics for chemistry_score:
Mean: 79.995
Median: 81.0
Mode: 94
Range: 50
Variance: 163.19297500000002
Standard deviation: 12.774700583575335


Descriptive statistics for biology_score:
Mean: 79.5815
Median: 81.0
Mode: 100
Range: 70
Variance: 188.20435775
Standard deviation: 13.718759337126663


Descriptive statistics for english_score:
Mean: 81.2775
Median: 83.0
Mode: 90
Range: 49
Variance: 144.57849375
Standard deviation: 12.024079746492037


Descriptive statistics for geography_score:
Mean: 80.888
Median: 81.0
Mode: 88
Range: 40
```

```
Variance: 135.368456
Standard deviation: 11.634795056209628


The subject with the highest variance in scores is: biology_score
With a variance of: 188.20435775
```

In [30]:

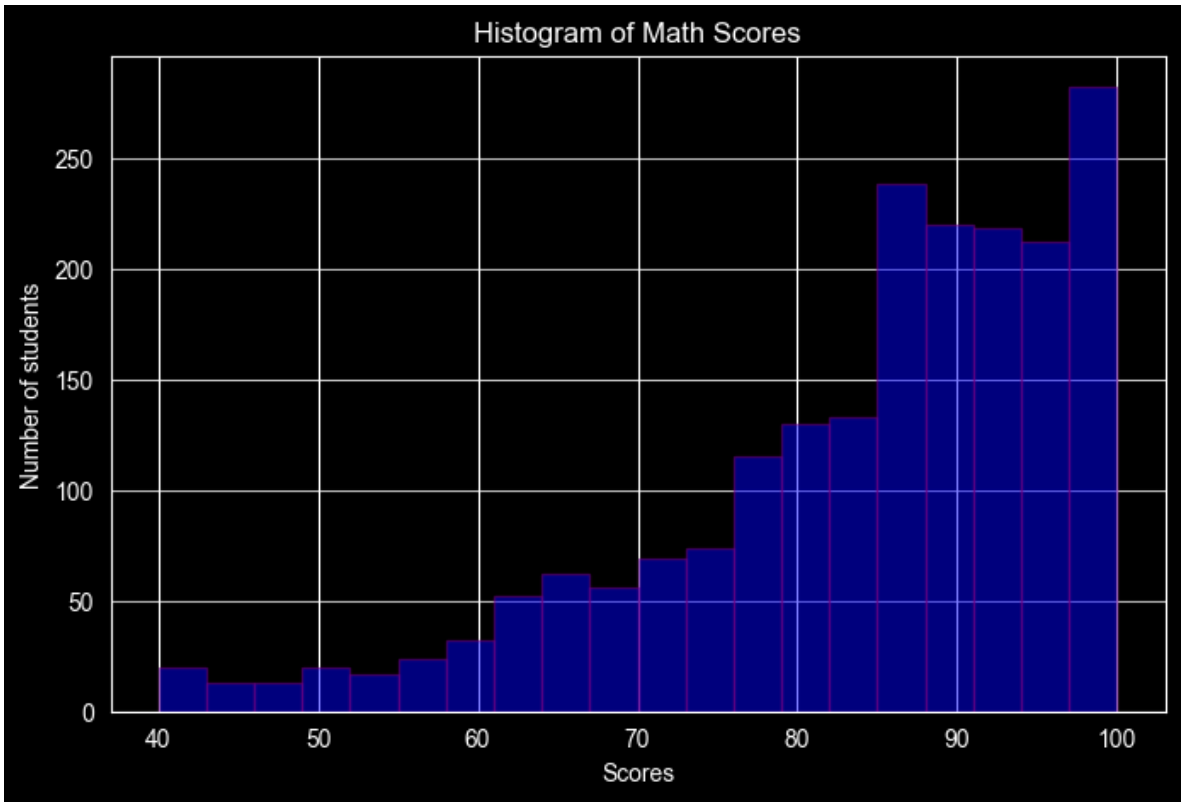
```
# Compare the descriptive statistics obtained from Pandas and NumPy

# This can be done by comparing the values obtained from both methods for each s

for i, subject in enumerate(desc_stats_dict['subject']):
    print("\n")

    print(f"Comparing descriptive statistics for {subject}:")

    print(f"Mean: Pandas = {summary_stats.loc['mean', subject]}, "
          f"NumPy = {desc_stats_dict['mean'][i]}")

    print(f"Median: Pandas = {summary_stats.loc['50%', subject]}, "
          f"NumPy = {desc_stats_dict['median'][i]}")

    print(f"Mode: Pandas = {df_scores[subject].mode().iloc[0]}, "
          f"NumPy = {desc_stats_dict['mode'][i]}")

    print(f"Range: Pandas = {summary_stats.loc['max', subject] - summary_stats.l
          f"NumPy = {desc_stats_dict['range'][i]}")

    print(f"Variance: Pandas = {df_scores[subject].var()}, "
          f"NumPy = {desc_stats_dict['variance'][i]}")

    print(f"Standard deviation: Pandas = {summary_stats.loc['std', subject]}, "
          f"NumPy = {desc_stats_dict['std'][i]}")

# The values obtained from both methods should match closely, confirming the con

# Variance and standard deviation values may differ slightly due to the default

# To ensure they match, we can specify ddof=1 in the NumPy calculations for vari
```

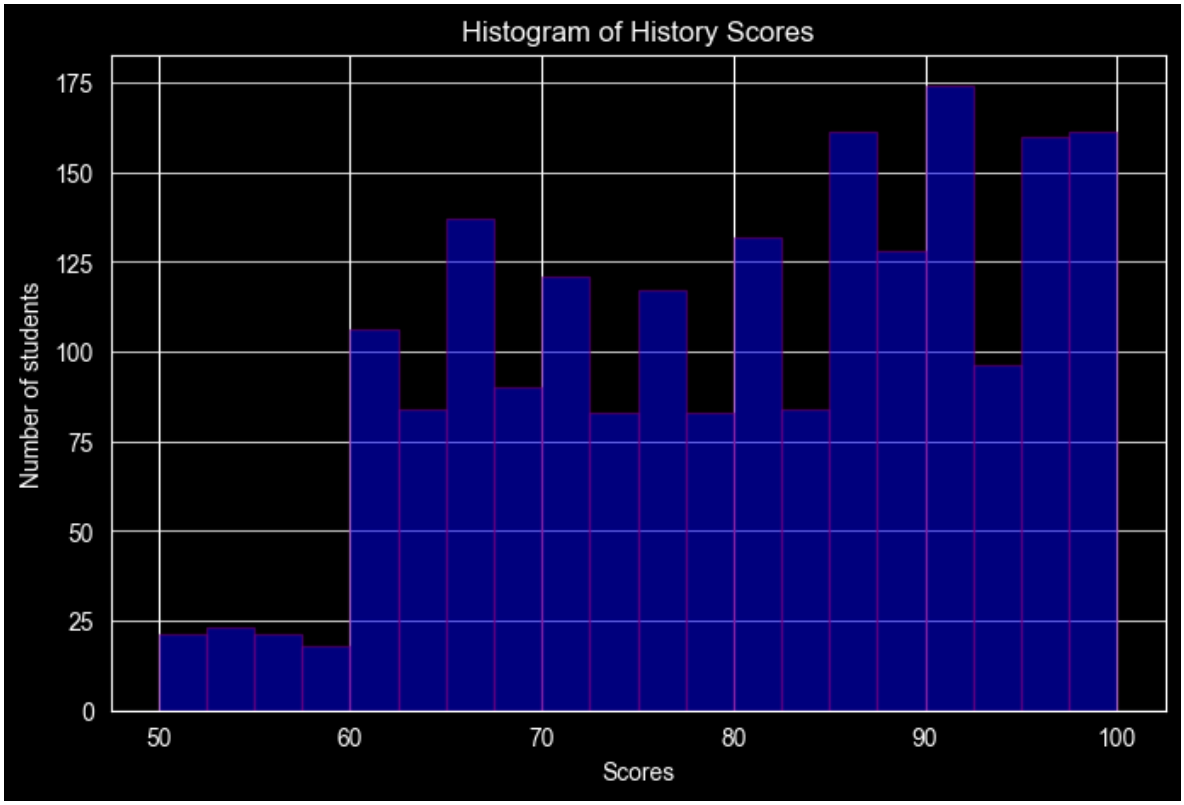
```
Comparing descriptive statistics for math_score:
Mean: Pandas = 83.452, NumPy = 83.452
Median: Pandas = 87.0, NumPy = 87.0
Mode: Pandas = 99, NumPy = 99
Range: Pandas = 60.0, NumPy = 60
Variance: Pandas = 174.89814507253627, NumPy = 174.810696
Standard deviation: Pandas = 13.22490624059529, NumPy = 13.22159960065347


Comparing descriptive statistics for history_score:
Mean: Pandas = 80.332, NumPy = 80.332
Median: Pandas = 82.0, NumPy = 82.0
Mode: Pandas = 88, NumPy = 88
Range: Pandas = 50.0, NumPy = 50
Variance: Pandas = 162.20687943971987, NumPy = 162.125776
Standard deviation: Pandas = 12.736046460331396, NumPy = 12.73286205061533


Comparing descriptive statistics for physics_score:
Mean: Pandas = 81.3365, NumPy = 81.3365
Median: Pandas = 83.0, NumPy = 83.0
Mode: Pandas = 96, NumPy = 96
Range: Pandas = 50.0, NumPy = 50
Variance: Pandas = 157.23788669334667, NumPy = 157.15926775
Standard deviation: Pandas = 12.539453205516844, NumPy = 12.536317950259559


Comparing descriptive statistics for chemistry_score:
Mean: Pandas = 79.995, NumPy = 79.995
Median: Pandas = 81.0, NumPy = 81.0
Mode: Pandas = 94, NumPy = 94
Range: Pandas = 50.0, NumPy = 50
Variance: Pandas = 163.27461230615307, NumPy = 163.19297500000002
Standard deviation: Pandas = 12.777895456848638, NumPy = 12.774700583575335


Comparing descriptive statistics for biology_score:
Mean: Pandas = 79.5815, NumPy = 79.5815
Median: Pandas = 81.0, NumPy = 81.0
Mode: Pandas = 100, NumPy = 100
Range: Pandas = 70.0, NumPy = 70
Variance: Pandas = 188.29850700350178, NumPy = 188.20435775
Standard deviation: Pandas = 13.722190313630758, NumPy = 13.718759337126663


Comparing descriptive statistics for english_score:
Mean: Pandas = 81.2775, NumPy = 81.2775
Median: Pandas = 83.0, NumPy = 83.0
Mode: Pandas = 90, NumPy = 90
Range: Pandas = 49.0, NumPy = 49
Variance: Pandas = 144.65081915957978, NumPy = 144.57849375
Standard deviation: Pandas = 12.027086894156032, NumPy = 12.024079746492037


Comparing descriptive statistics for geography_score:
Mean: Pandas = 80.888, NumPy = 80.888
Median: Pandas = 81.0, NumPy = 81.0
Mode: Pandas = 88, NumPy = 88
Range: Pandas = 40.0, NumPy = 40
```

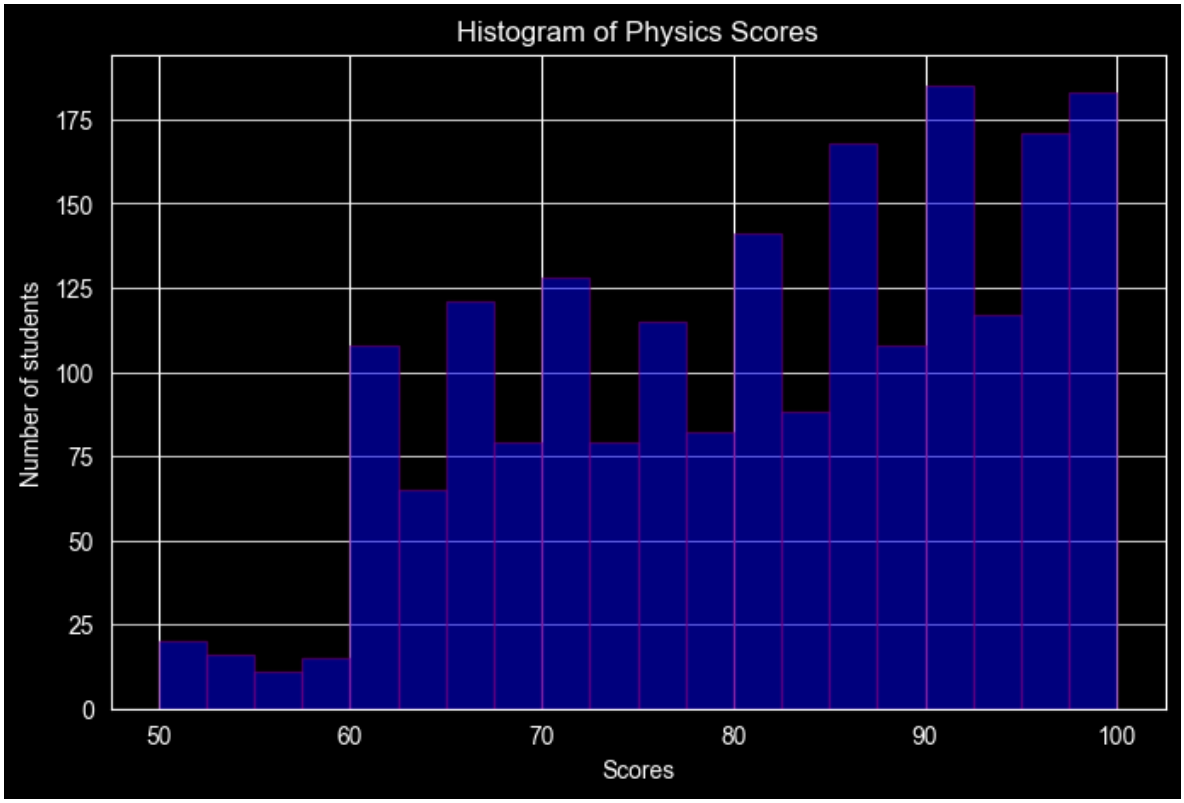
```
Variance: Pandas = 135.43617408704353, NumPy = 135.368456
Standard deviation: Pandas = 11.6377048461904, NumPy = 11.634795056209628
```

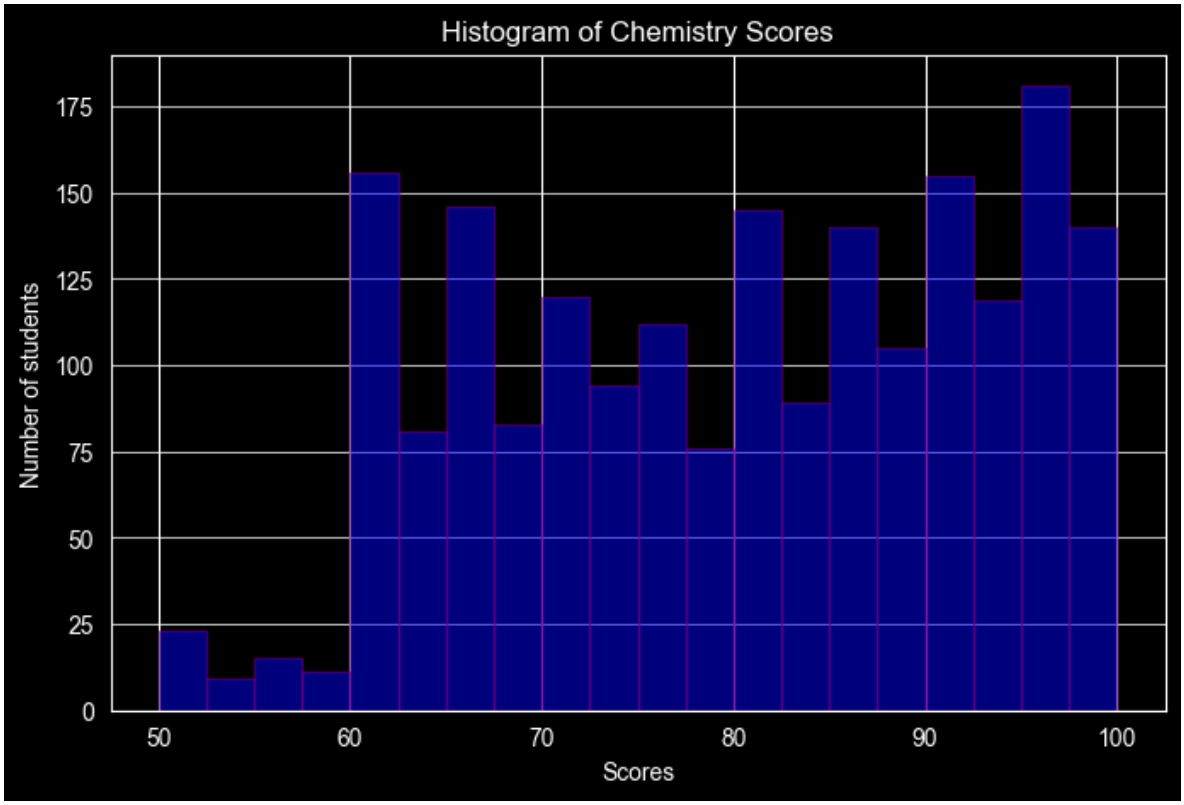
Task 5: Data visualisation

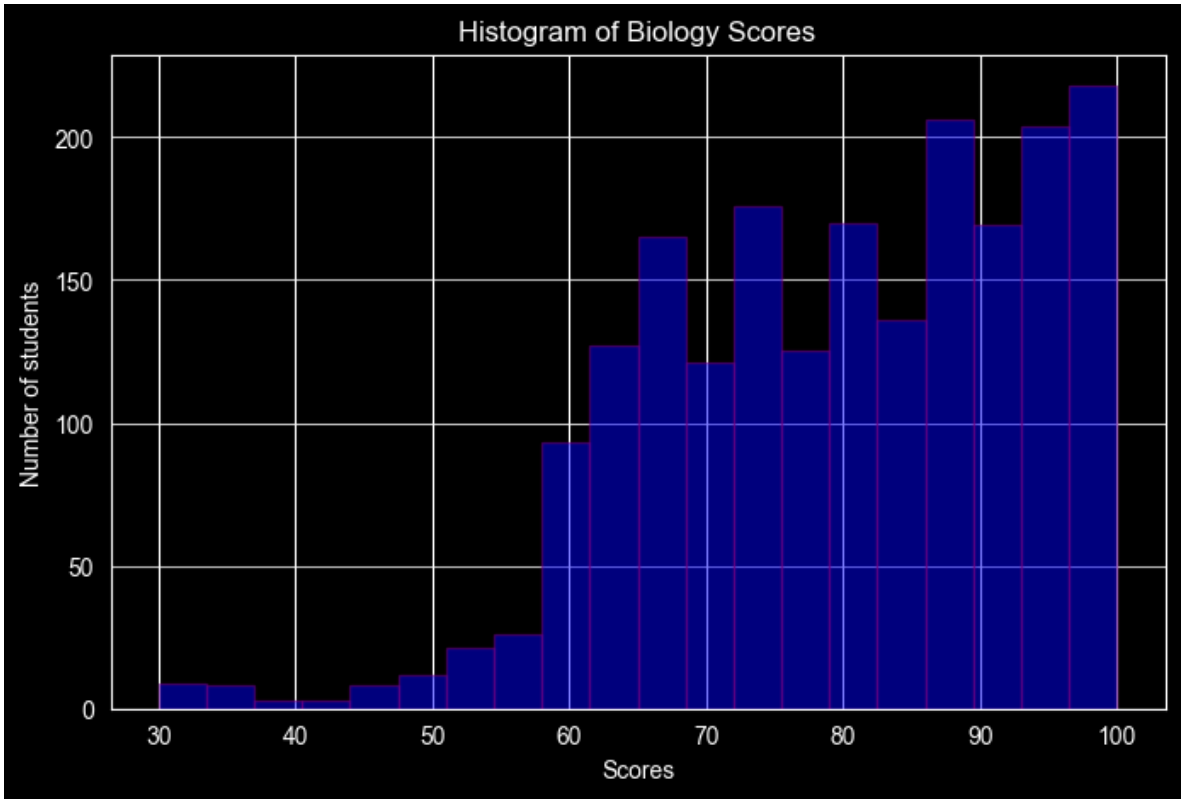
In [31]:
```
# Plot histograms for each subject to visualise the score distributions
subjects = df_scores.columns
for subject in subjects:
    plt.figure(figsize=(8, 5))
    plt.hist(df_scores[subject], alpha=0.5, bins=20, label=subject, edgecolor='p
    subject_string = str.split(subject, "_")[0].capitalize()
    plt.title(f'Histogram of {subject_string} Scores')
    plt.xlabel('Scores')
    plt.ylabel('Number of students')
    plt.grid(axis='y', alpha=0.75)
    plt.show()
```

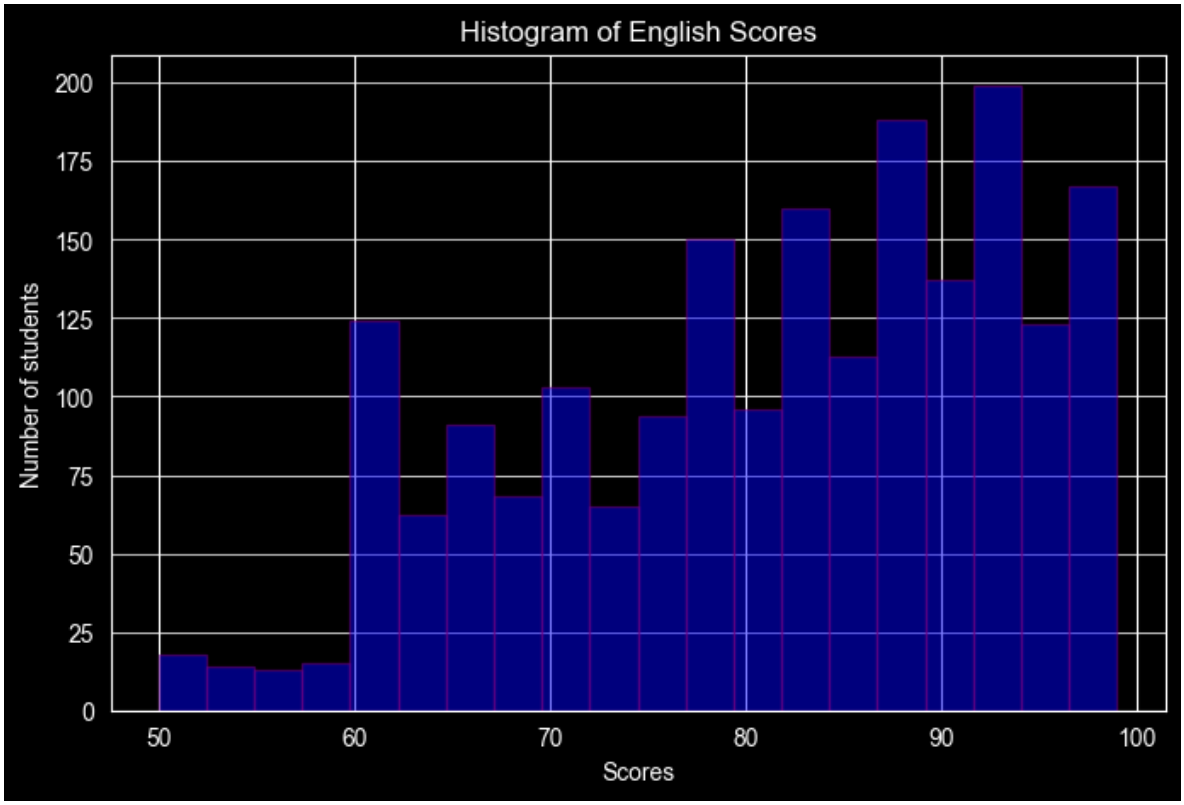


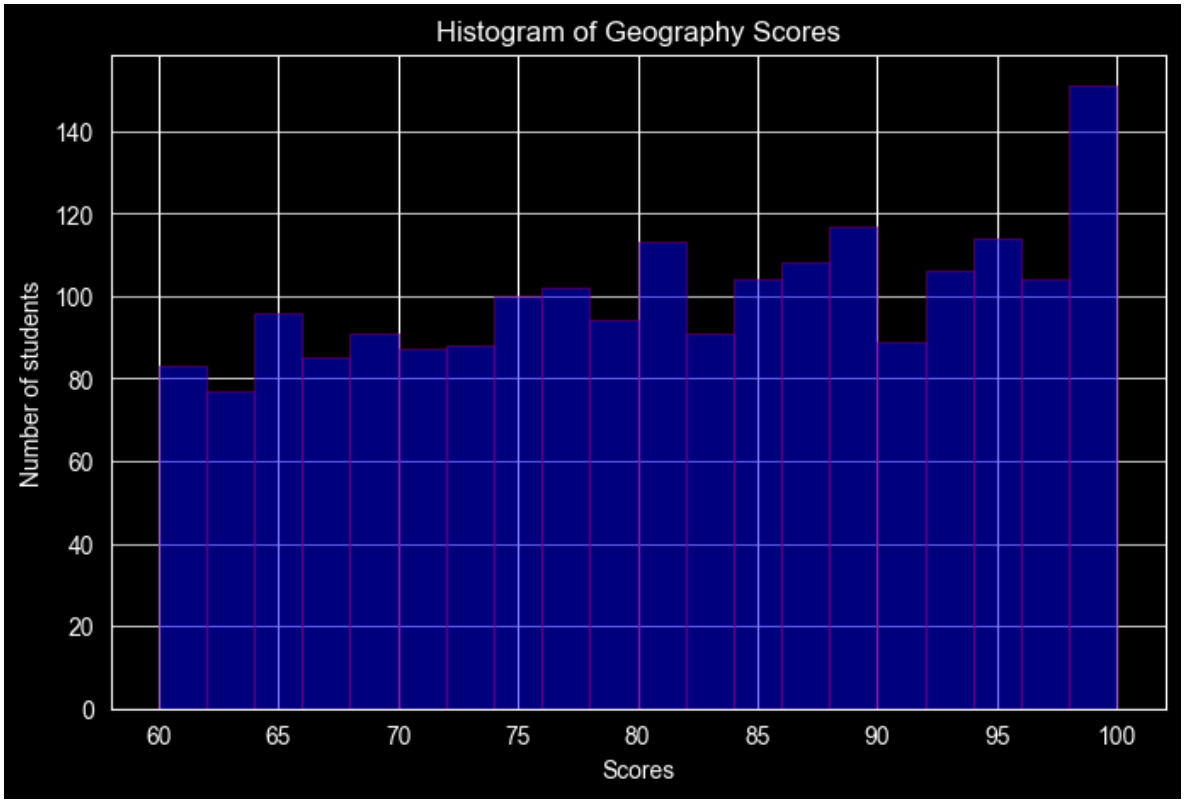

Histogram of History Scores
Number of students
Scores
0
25
50
75
100
125
150
175
50
60
70
80
90
100


Histogram of Physics Scores
Number of students
Scores
0
25
50
75
100
125
150
175
50
60
70
80
90
100


Histogram of Chemistry Scores
Number of students
0
25
50
75
100
125
150
175
50
60
70
80
90
100
Scores

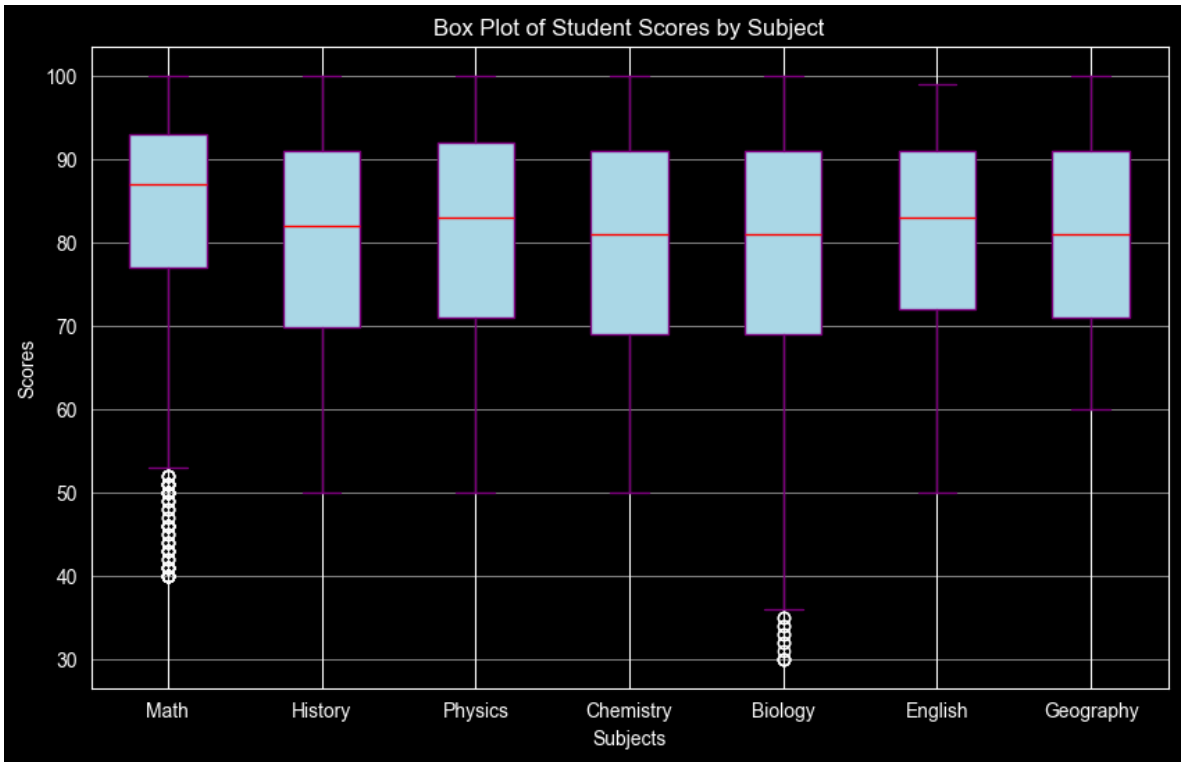

Histogram of Biology Scores
Number of students
0
50
100
150
200
30
40
50
60
70
80
90
100
Scores





In [32]:
```
# Create a box plot to compare the score distributions across the subjects
subjects = df_scores.columns
pretty_subjects = [str.split(subject, "_")[0].capitalize() for subject in subjec
plt.figure(figsize=(10, 6))
plt.boxplot([df_scores[subject] for subject in subjects],
            tick_labels=pretty_subjects,
            patch_artist=True,
            boxprops=dict(facecolor='lightblue', color='purple'),
            medianprops=dict(color='red'), whiskerprops=dict(color='purple'), ca
plt.title('Box Plot of Student Scores by Subject')
plt.xlabel('Subjects')
plt.ylabel('Scores')
```

```
plt.grid(axis='y', alpha=0.5)
plt.show()
```



Task 6: Correlation analysis between subjects

In [33]:
```
# Calculate the correlation matrix to understand the relationships between score

# This can be done using the corr() method of Pandas DataFrame
pandas_correlation_matrix = df_scores.corr()
pandas_correlation_matrix = pandas_correlation_matrix.round(8)
print(pandas_correlation_matrix)
```

```
                math_score  history_score  physics_score  chemistry_score  \
math_score        1.000000       0.147247       0.115719         0.127131
history_score     0.147247       1.000000       0.048478         0.121498
physics_score     0.115719       0.048478       1.000000         0.126163
chemistry_score   0.127131       0.121498       0.126163         1.000000
biology_score     0.081298       0.088502       0.132280         0.119992
english_score     0.134831       0.147193       0.054314         0.068341
geography_score   0.049672       0.065751       0.103126         0.065430

                biology_score  english_score  geography_score
math_score           0.081298       0.134831         0.049672
history_score        0.088502       0.147193         0.065751
physics_score        0.132280       0.054314         0.103126
chemistry_score      0.119992       0.068341         0.065430
biology_score        1.000000       0.074227         0.106526
english_score        0.074227       1.000000         0.072250
geography_score      0.106526       0.072250         1.000000
```

In [34]:
```
# This can also be done using the np.corrcoef() method of NumPy on the NumPy arr
numpy_correlation_matrix = np.corrcoef(scores_array, rowvar=False)
numpy_correlation_matrix = np.round(numpy_correlation_matrix, 8)
numpy_correlation_matrix
```

Out[34]:
```
array([[1.        , 0.14724747, 0.11571874, 0.12713149, 0.08129806,
        0.13483078, 0.04967233],
       [0.14724747, 1.        , 0.04847843, 0.12149786, 0.08850197,
        0.14719288, 0.06575135],
       [0.11571874, 0.04847843, 1.        , 0.1261628 , 0.13227971,
        0.0543137 , 0.10312592],
       [0.12713149, 0.12149786, 0.1261628 , 1.        , 0.11999168,
        0.06834058, 0.06543008],
       [0.08129806, 0.08850197, 0.13227971, 0.11999168, 1.        ,
        0.07422691, 0.10652591],
       [0.13483078, 0.14719288, 0.0543137 , 0.06834058, 0.07422691,
        1.        , 0.07224979],
       [0.04967233, 0.06575135, 0.10312592, 0.06543008, 0.10652591,
        0.07224979, 1.        ]])
```

In [35]:
```python
# Print the correlation matrix to understand the relationships between scores in
subjects = df_scores.columns

for i in range(len(subjects)):
    for j in range(i + 1, len(subjects)):
        print(f"{subjects[i]} ↔ {subjects[j]}:"
              f" {numpy_correlation_matrix[i, j]:.8f}")
```

```
math_score ↔ history_score: 0.14724747
math_score ↔ physics_score: 0.11571874
math_score ↔ chemistry_score: 0.12713149
math_score ↔ biology_score: 0.08129806
math_score ↔ english_score: 0.13483078
math_score ↔ geography_score: 0.04967233
history_score ↔ physics_score: 0.04847843
history_score ↔ chemistry_score: 0.12149786
history_score ↔ biology_score: 0.08850197
history_score ↔ english_score: 0.14719288
history_score ↔ geography_score: 0.06575135
physics_score ↔ chemistry_score: 0.12616280
physics_score ↔ biology_score: 0.13227971
physics_score ↔ english_score: 0.05431370
physics_score ↔ geography_score: 0.10312592
chemistry_score ↔ biology_score: 0.11999168
chemistry_score ↔ english_score: 0.06834058
chemistry_score ↔ geography_score: 0.06543008
biology_score ↔ english_score: 0.07422691
biology_score ↔ geography_score: 0.10652591
english_score ↔ geography_score: 0.07224979
```